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Vehicle docking structure with a solar panel

Abstract

In one embodiment, an apparatus for docking vehicles includes a housing structure comprising two side portions and a top portion connecting the two side portions, the two side portions and the top portion forming a cavity for receiving a portion of a vehicle, an electrically-powered locking assembly, carried by the housing structure, for securing the housing structure to the vehicle in response to the portion of the vehicle being received by the cavity of the housing structure, an electrically-powered communication device, coupled to the housing structure, for enabling data communication over a network, and a solar panel, carried by an external surface of one of the side portions of the housing structure, for independently converting light received by the solar panel into electrical energy.

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Background/Summary

PRIORITY [0001] This application is a continuation under 35 U.S.C. § 120 of U.S. patent application Ser. No. 17/565,269, filed 29 Dec. 2021, which is incorporated herein by reference.

BACKGROUND

[0002] Traditional configurations for docking stations rely on a centralized power source (e.g., electrical grid, battery) and a kiosk that utilizes a centralized communication module to support its docking mechanism. There is a reliability concern with such centralized configurations since they have a single point of failure that could render unavailable every rideable/vehicle docked at the station (e.g., bikes, scooters, and etc.). Moreover, due to relying upon the centralized power source, the initial costs involved with the centralized configuration make it only feasible to deploy docking stations at heavily populated areas and less feasible at less populated areas where only a few docks may be required. Furthermore, docking stations connected to the electrical grid require higher deployment costs and are less flexible on where they can be deployed (e.g., only where the electrical grid is available), and docking stations supported by a battery require additional labor costs, for example, due to the necessary battery swaps every so often (e.g., on a monthly basis). Accordingly, there is a need for a new dock or a docking station that is more reliable and flexible in terms of where it can be deployed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 illustrates an example docking station with a kiosk.

[0004] FIG. 2 illustrates example docks configured with a solar panel, according to some embodiments.

[0005] FIG. 3 illustrates another example of docks configured with a solar panel, according to some embodiments.

[0006] FIGS. 4A-4F illustrate an example dock and its components, according to some embodiments.

[0007] FIG. 5 illustrates an example of electrical components of a dock, according to some embodiments.

[0008] FIG. 6 illustrates an example docking station, according to some embodiments.

[0009] FIG. 7 illustrates an example docking station connected to an external power source, according to some embodiments.

[0010] FIG. 8 illustrates an example of electrical components of a docking station connected to an external power source, according to some embodiments.

[0011] FIG. 9 illustrates a map of example docking stations installed in a region, according to some embodiments.

[0012] FIG. 10 illustrates an example method for communicating with individual docks at a docking station, according to some embodiments.

[0013] FIG. 11 illustrates an example computer system, according to some embodiments.

DESCRIPTION OF EXAMPLE EMBODIMENTS

[0014] In the following description, various embodiments will be described. For purposes of explanation, specific configurations and details are set forth in order to provide a thorough understanding of the embodiments. However, it will also be apparent to one skilled in the art that the embodiments may be practiced without the specific details. Furthermore, well-known features may be omitted or simplified in order not to obscure the embodiment being described. In addition, the embodiments disclosed herein are only examples, and the scope of this disclosure is not limited to them. Particular embodiments may include all, some, or none of the components, elements, features, functions, operations, or steps of the embodiments disclosed above. Embodiments according to the invention are in particular disclosed in the attached claims directed to a method, a storage medium, a system and a computer program product, wherein any feature mentioned in one claim category, e.g., method, can be claimed in another claim category, e.g., system, as well. The dependencies or references back in the attached claims are chosen for formal reasons only. However, any subject matter resulting from a deliberate reference back to any previous claims (in particular multiple dependencies) can be claimed as well, so that any combination of claims and the features thereof are disclosed and can be claimed regardless of the dependencies chosen in the attached claims. The subject-matter which can be claimed comprises not only the combinations of features as set out in the attached claims but also any other combination of features in the claims, wherein each feature mentioned in the claims can be combined with any other feature or combination of other features in the claims. Furthermore, any of the embodiments and features described or depicted herein can be claimed in a separate claim and/or in any combination with any embodiment or feature described or depicted herein or with any of the features of the attached claims.

[0015] The architecture of a self-sufficient dock disclosed herein provides a technical solution to the issues involved with docking stations relying on a centralized power source (e.g., electrical grid, battery) and a centralized communication module. Embodiments described herein disclose a self-sufficient, independently-operated dock with improved reliability, low operation costs, and flexibility to be deployed in both high- or low-density areas, all without requiring a kiosk, a centralized power source, or underground power and communication infrastructure. As used herein, the term “self-sufficient” may refer to a dock capable of independently generating sufficient power to power its various docking functionalities without requiring or using power generated by other docks or a power source external to the dock. In some embodiments, the various docking functionalities include locking the rideable at the dock, unlocking the rideable from the dock, communicating with a transportation management system (e.g., to indicate the status of a dock with respect to whether a rideable is docked to the dock and whether a rideable is locked or unlocked to the dock), and for locking out the dock to reserve the dock for a rideable, but the docking functionalities may not include charging a rideable. In some embodiments, the various docking functionalities may also include charging the rideable. Indeed, the docking functionalities performed by the dock require a significant amount of power. A dock may include a “housing structure” that may take the form of a bollard. Each dock is configured with one or more solar panels, battery, charging system, locking mechanism, and communication system, allowing it to be self-sufficient and independently deployable (even in low-density areas). The solar panels may be configured to draw power from ambient or diffuse light and do not require direct sunlight to supply the necessary power to operate the various docking functionalities. The solar panels may be also oriented vertically relative to the ground or the platform of the dock to allow the space above the dock to be freed and occupied by portions of a bike, such as its basket. In some examples, the solar panels may be diagonally oriented (i.e., any angle range greater than 0 degrees to less than 90 degrees between the light-absorbing surface of the solar panel relative to the surface of the ground or the dock platform). Given that the solar panels do not require direct sunlight, the docks can be positioned close to each other or close to other non-docking structures that could potentially block

direct sunlight, as may be often present in high-density areas. The vertical orientation or diagonal orientation is also beneficial since it provides natural resistance to dust buildup over the solar panels, or in high latitude markets, snow buildup. Each dock may also be configured to be connected to an external power source to provide the capability to charge electrical vehicles (e.g., bikes, scooters, etc.) although it should be noted in some examples the self-sufficient docking stations described herein may generate sufficient power to charge electrical vehicles

[0016] FIG. 1 illustrates an example of a traditional docking station **100** with a kiosk **105**.

Traditional docking stations may comprise a kiosk and multiple docks, each dock **122** comprising a docking mechanism for securing a rideable **136**. A rideable **136** is a transportation vehicle that a user could ride on to travel from point A to point B. Examples of a rideable include a bike, scooter, or other forms of a rideable vehicle. Traditional docking stations rely on a centralized power source (e.g., electrical grid, battery) to support its docking mechanism and a kiosk. A kiosk **105** is an interactive terminal that provides users with access to, and information about, rideables **136** that are docked at a station. Traditional docking stations **100** are centralized in that the kiosk **105** and docks **122** are powered by a single power source (e.g., electrical grid or a battery). The centralized configuration has a single point of failure that, if failed, could render unavailable every rideable docked at the station, for example, when the electrical grid is down, if a station battery depletes, or the communication module of a kiosk malfunctions. Moreover, the deployment and maintenance costs involved with the centralized configuration make it only feasible to deploy such docking stations at heavily populated areas and unfeasible at less populated areas where only a few docks may be required due to low use of rideables. For example, it may not be economically efficient to deploy only a single dock in a traditional docking station with a kiosk due to the cost of installing or maintaining the kiosk. Indeed, it may not be economically efficient to deploy a single dock because installing a traditional dock **122** on a street requires city permits to dig underground to install long, underground telecommunication lines and electrical power cables connecting to an electrical grid in order to support the communication and power needs of the traditional dock **122**. In some instances, the traditional dock **122** cannot be deployed in areas where there are underground restrictions (e.g., sewage lines). In other words, the deployment of the traditional dock **122** may be limited by underground restrictions. Furthermore, docking stations that are connected to the electrical grid require additional components and labor for interfacing with the grid and thus are less flexible on where they can be deployed (e.g., only where the electrical grid is available). Docking stations supported by a station battery also require additional labor, for example, for swapping out the station battery every so often (e.g., on a monthly basis). Embodiments disclosed herein provides a new type of a dock that could operate independently from a kiosk and does not rely on an external power source such as an electrical grid to operate its various docking functionalities.

[0017] FIG. 2 illustrates examples of self-sufficiently powered, independently-operated docks **200** with improved reliability, low operation costs, and flexibility to be deployed in both high- or low-dense areas. In contrast to a traditional dock **122**, a dock **200** may be configured with a solar panel **211** that is capable of drawing sufficient power from diffused, ambient, albedo, or reflected irradiance/light (rather than direct sunlight) to operate the electrically-powered components within the dock **200**. As illustrated in FIG. 2, the solar panel **211** may be oriented vertically relative to the ground such that the solar panel **211** may not be positioned to receive direct sunlight during daytime. Additionally, orienting the solar panel **211** in a vertical orientation may minimize the footprint of each dock **200**; thereby, maximizing the number of docks **200** that may be installed in an area with an overall small footprint. For example, in an urban downtown area such as a corner of a busy intersection in front of a storefront, it is economically efficient to deploy even a handful of docks **200** at the corner because each of the docks **200** is self-sufficient and its primary power systems and communication systems are included within its housing structure and above ground. In contrast, a traditional dock **122** may require city permits to dig underground to install underground

telecommunication lines and electrical power cables connecting to an electrical grid that are necessary to power and communicate with the traditional dock **122**.

[0018] FIG. **3** illustrates an example of a dock **200** that is receiving diffused reflection **261** of an incident light **253** (e.g., sunlight, ambient light, etc.), which diffused reflection **261** has been reflected off of an object in an external environment (e.g., a structure). The self-sufficient docks **200** that are disclosed herein are able to draw sufficient power from such indirect light to power its internal docking components (e.g., locking mechanism, wireless communications, etc.). The self-sufficient docks **200** that are disclosed herein are able to operate based on indirect light (e.g., diffused, ambient, albedo, or reflected irradiance/light), and thus are much more flexible on where they can be deployed (e.g., even where direct sunlight is not available). The self-sufficient docks **200** are also flexible in how they can be deployed, for example, closely next to each other (e.g., distance of 1 foot apart), as shown in FIG. **3**. Such a configuration addresses a deficiency in a dock configuration that requires direct sunlight to power docking components since docks that are too close to each other may block light from reaching each others' solar panels. Although docks that are exposed to direct sunlight may receive a superficial amount of indirect sunlight, the solar panels and related power systems of these docks are incapable of absorbing and converting a sufficient amount of indirect light during the daytime in order to provide necessary docking functions throughout the evening and/or during an extended period where there is a lack of sunlight. Another benefit to having vertically orientated solar panels is that the vertically orientated solar panels will not be covered by a portion of a bike **136**, for example, a basket that is mounted in front of the bike. [0019] FIGS. **4A-4G** illustrate an example dock **200** with vertically oriented solar panels **211**. In particular embodiments, as illustrated in FIG. **4A**, a dock **200** is comprised of a bollard structure **420**, a vertically oriented solar panel **211**, a cassette assembly **430** comprising battery, communication module, and locking assembly. In some embodiments, the cassette assembly **430** may include a user interface panel for the rider to execute functions at the dock **200**. In particular embodiments, a dock **200** may be configured with a single solar panel **211**, or two solar panels **211** (e.g., one on each side of the bollard structure).

[0020] FIG. **4B** illustrates an exploded view of an example dock **200**. In particular embodiments, a dock **200** may be comprised of a bollard structure **420** with a cavity **424** for receiving a rideable, a cassette assembly **430**, a housing **431** for the cassette assembly, a solar panel **211**, and a base **461**. In particular embodiments, a base **462** may be configured as a rod that extends downwards into the ground. In particular embodiments, a base **463** could be installed on a raised platform. In particular embodiments, a solar panel **211** may comprise a light-absorbing surface **212** that includes photovoltaic cells, which are electrical devices that convert light energy into electricity by the photovoltaic effect.

[0021] FIG. **4C** illustrates another exploded view of an example dock **200**. In particular embodiments, a dock **200** may comprise an inner component/structure **471** having walls that define a cavity for receiving and securing a rideable therein. In particular embodiments, the base **461** of the dock **200** may be configured on top of a platform **481**, under which runs electrical wiring that may be connected to the dock **200**. As shown in FIG. **4C**, a solar panel **211** may be configured on one side, or both sides of the bollard structure **420**. If a solar panel is not configured on a side of the bollard structure **420**, a cover panel may be configured in its place instead. The base **461** may provide a supportive structure for supporting the dock **200** above ground. In some examples, the base **461** may lack holes or cavities such as to exclude any internal power cables and communication lines from passing between the dock **200** and underground. In this manner, the base **461** may be self-sufficient in satisfying its own power and communication requirements.

[0022] FIG. **4D** illustrates an example bollard structure **420** with a linear aluminum profile. In particular embodiments, a bollard structure **420** may comprise a front cavity **425** and a back cavity **426** for receiving a rideable. For example, if a rideable is a bike, its front wheel would go through the front cavity **425** and may extend out through the back cavity **426**. If a rideable is a scooter, a

portion of the scooter may go into the front cavity **425** but may not necessarily extend out through the back cavity **426**. In particular embodiments, a dock **200** may only comprise a front cavity **425** (without a back cavity **426**). In particular embodiments, a bollard structure **420** may have a linear aluminum profile that is constructed using a single extrusion die. In particular embodiments, a solar panel may be fitted in the side opening **427** of a bollard structure **420**.

[0023] FIG. **4E** illustrates an example bollard structure **420** with a separated aluminum profile. In particular embodiments, a bollard structure **420** may be comprised of two separate components, in contrast to the single aluminum profile shown in the embodiment of FIG. **4D**. In some examples, the bollard structure **420** may be comprised of metal, composite, plastic, or a combination thereof.

[0024] FIG. **4F** illustrates an example cassette **430** of a dock **200**. In particular embodiments, a cassette **430** carries a battery **431** for storing the energy harvested from solar panels, a communication module **432** for communicating wirelessly with external devices or systems (e.g., bike transportation management system), a locking mechanism **564**, and a sensor for detecting the docking status of a vehicle and/or the dock **200**. The sensor (not illustrated in FIG. **4F**) may be installed within, or next to, the locking mechanism **564**. In particular embodiments, the battery **431** may be a lithium battery, a single or double AA battery, or a single or double **18650** cell battery. In particular embodiments, a vehicle may comprise a locking member that is configured to be received by a locking mechanism **564**. When a locking member is received by a locking mechanism **564**, a locking mechanism **564** may mechanically lock the vehicle associated with the locking member to the dock **200**. This locking process may occur automatically and mechanically based solely on the act of a vehicle (e.g., the locking member) being pushed into the locking mechanism **564**, that is, without requiring any power (e.g., electrical energy). In some examples, the locking process may utilize power (as sourced from indirect sunlight) to lock the vehicle to the dock **200** using the locking mechanism **564**. The unlocking process may require utilizing power. In particular embodiments, the unlocking process may begin when a dock **200**'s communication module **432** receives instructions or authentication message from a client device or a server to unlock a vehicle that is docked and locked to a dock **200**. The locking mechanism **564** may then utilize power stored in a battery **431** to unlock the vehicle from the dock **200**. At this point, the vehicle may be unlocked but still be docked at the dock **200**. In particular embodiments, a locking mechanism **564** may comprise a sensor that is configured to detect whether a vehicle is docked to a dock **200** and whether a rideable is locked or unlocked to the dock **200**. For a particular dock **200**, a sensor may be configured to detect any of the following statuses: (1) no vehicle is docked, (2) a vehicle is docked and locked, and (3) a vehicle is docked but unlocked. In one example, if no vehicle is docked at the respective dock **200** based on the sensor, the respective dock may be configured to indicate that it is unavailable for use. In another example, if a vehicle is docked at the respective dock **200** based on the sensor, then the respective dock may be configured to indicate that it is occupied.

[0025] FIG. **5** illustrates example electrical components of a dock **200**. In particular embodiments, a dock **200** may comprise a solar panel **211** that is configured to convert light/irradiance into electrical energy, an energy harvester **520** that is configured to charge a battery **530** using the converted energy, and a load **560** (e.g., communication module **562**, locking mechanism **564**, and sensor **567**). In particular embodiments, the communication module **562** may comprise a cellular modem that enables wireless communication with a cell tower. The communication module **562** may also comprise other communication modules (near-field communication, bluetooth), for example, to communicate with client devices. The communication module **562** may also be configured to communicate with a bike/scooter transportation management system to receive instructions and to facilitate communication of the respective status of one or more docks to the transportation management system. For example, the transportation management system may wirelessly send instructions to a communication module **562** of a particular dock for the purpose of preemptively reserving that dock for a particular rider during a predetermined time of the day such

that the particular dock is unavailable for use by other riders during the predetermined time of the day.

[0026] FIG. 6 illustrates an example docking station with two self-sufficient docks **200** that demonstrates the flexibility of how and where such self-sufficient docks **200** can be deployed. The self-sufficient docks **200** disclosed herein are able to independently generate its own power and independently communicate with external devices/systems, and thus are remarkably flexible on where and how they can be deployed. In contrast, traditional docking stations relying on a centralized power source are not so flexible because they require additional permitting and infrastructure (e.g., to connect to the power grid, installing a kiosk), which often translates to higher installation costs. Traditional docking stations also require a much bigger form factor, for example, because they require an external kiosk and/or electrical connections for connecting to the power grid. With the higher costs and bigger form factor, traditional docking stations are often difficult to install in high-density areas, such as a city downtown, where space is limited. The self-sufficient docks **200** disclosed herein provide a solution to those issues. Given that the solar panel, wireless communication module, and locking mechanism are all housed in each of the docks **200**, each of these docks require a much smaller form factor and can be independently deployed in a much smaller area than the traditional docking stations. With respect to economic feasibility, there is no minimum number of self-sufficient docks **200** that are required for each docking station. Meaning, at any particular location, even a single self-sufficient dock may be installed, or alternatively as many as needed (e.g., 10, 25, 50 or more). The flexibility in deployment allows the self-sufficient docks **200** to be installed in a high-density area, such as a curb of a busy downtown intersection, or otherwise an area with limited space. FIG. 6 illustrates an example of how self-sufficient docks **200** can be flexibly deployed, where just two self-sufficient docks **200** are installed between the limited space between two trees. Additionally, the self-sufficient docks **200** may also be installed in areas where the electrical grid is not available since each dock **200** can independently generate its own power to operate its docking functionalities.

[0027] FIG. 7 illustrates an example diagram of a docking station that is connected to an external power source **750**, such as an electrical grid. In particular embodiments, the dock station may include a solar panel **211** for providing sufficient power to independently operate the docking functionalities of each respective dock station, as described herein. In particular embodiments, a docking station that is connected to an electrical grid may comprise some docks that are connected to the grid and some docks that are not. For example, a docking station may comprise two docks that are connected to an electrical grid and ten other docks that are not connected to the grid. In such embodiments, docks that are connected to an electrical grid may be configured to route the power from the grid to an electric rideable **136** docked onto the docks, for example, to recharge a battery coupled to the electric rideable **136**.

[0028] FIG. 8 illustrates example electrical components of a docking station that is connected to an external power source, such as an electrical grid. In particular embodiments, a docking station may comprise an AC/DC bus supply **823** that interfaces between the AC electrical grid and a DC bus of the docking station. In particular embodiments, self-sufficient docks **200** disclosed herein may be configured to rely on an external power source to operate its electrically-powered components rather than relying on a solar panel. For example, FIG. 8 illustrates an example dock **855** that has been configured substantially similar to a self-sufficient dock **200** described above but without a solar panel, and instead, configured with a charger **802** for charging an electric rideable, a converter **801** for providing power to an energy harvester **520**, and a battery **530** for storing energy received from the energy harvester **520**. FIG. 8 illustrates another example dock **845** that is substantially similar to dock **855** but without a solar panel, energy harvester, and battery, and instead, the dock **845** is configured to allow the converter **801** to directly provide power to the load **560**. In some embodiments, self-sufficient docks **200** disclosed herein may be configured with solar panels and connected to an external power source. In such embodiments, the solar panels may be configured to

work in parallel with the external power source, or as a back-up power source to the external power source. In other embodiments, the solar panels may be configured to supply power to a dock's internal electrically-powered components while using the external power source to charge electric rideable. In such embodiments, a dock may comprise a charger **802** but not a converter **801**. In some embodiments, the dock may also utilize another energy harvesting mechanism besides a solar panel for capturing solar energy. For example, the dock may capture a renewable energy source (e.g., wind energy, hydroelectric, etc.) using an energy harvesting mechanism carried by the housing structure of the dock to independently generate sufficient energy to power the various docking functionalities.

[0029] FIG. **9** illustrates a diagram that shows various docking stations installed at different locations in a region. In particular embodiments, docking stations with charging capability **905** (for charging electric rideables) may be installed at certain locations in a region and docking stations without charging capability **980** may be installed at other locations in the region. For example, docking stations with charging capability **905** may be installed at locations with access to an electrical grid. Docking stations without charging capability **980** may be installed at locations without access to an electrical grid but may also be installed at locations with access to the grid. In particular embodiments, docking stations with charging capability **905** may comprise some docks that are connected to an external power source and other docks that are not connected to the external power source. For example, in a low-density area with low foot-traffic, a docking station with charging capability **905** may comprise a single dock that is connected to the electrical grid and three other docks that are not connected to the electrical grid. As another example, in high-density areas with high foot-traffic, a docking station with charging capability **905** may comprise five docks that are connected to the electrical grid and twenty-five other docks that are not connected to the electrical grid. In particular embodiments, only the docks **200** that are connected to an electrical grid may be configured with a charger for charging an electric rideable. In particular embodiments, foot traffic patterns at various locations may be determined based on GPS location tracking of user devices or based on a measurement of a number of rideables that are rented from, or docked to, particular docking stations over a particular period of time. In particular embodiments, different number of docks may be installed in various locations based on user demand, which may be determined based on (1) historical data indicating how often rideables are rented from a particular docking station, (2) based on foot traffic patterns, or (3) based on location tracking. For example, each of docking stations **980** shows that it comprises three docks (e.g., represented by three dots) and docking station **991** shows that it comprises four docks. As another example, given that a hospital is a location commonly visited by users, FIG. **9** illustrates that docking-and-charging stations **915** that are installed by the hospital comprise more docks than the docking-and-charging stations **905** that are scattered around shops, restaurants, and bars.

[0030] As illustrated in FIG. **9**, the geographic region includes a combination of both docking-and-charging station **905** and non-charging stations (e.g., docking station **980** and docking station **991**). Indeed, it may be costly to implement docking-and-charging station **905** at every location in the geographic region that is eligible for a docking station due to the high cost in building and maintaining the charging infrastructure needed to connect each of the docking-and-charging station **905** to an electrical grid. As a result, there can be a point of diminishing returns if charging stations are implemented in every location in the geographic region that makes it infeasible to do so. FIG. **9** illustrates that the docking-and-charging station **905** is strategically positioned in regions that have a large amount of foot traffic (e.g., close to shops, restaurants, bars, hospital, etc.) and are also within travel distance to the non-charging stations (e.g., docking station **980** and docking station **991**) such that an electrically-powered rideable undocked by a rider from the docking station **980** has ample charge to ride to any one of the docking-and-charging stations **905**.

[0031] Although FIG. **9** illustrates that there are more docking-and-charging stations **905** than non-charging stations (e.g., docking station **980** and docking station **991**), it should be noted that any

amount of distribution between charging and non-charging docking stations may be utilized in the geographic region in order to meet the needs of the geographic area. For example, if the geographic area has a centrally located downtown city area with shops, restaurants, bars, entertainment venues that is surrounded by suburban areas, then the downtown city area may include the majority of the charging stations while the suburban areas may include the majority of the non-charging stations. In this manner, the high cost in building and maintaining the charging infrastructure needed to connect each of the charging stations to an electrical grid is balanced by high ridership in the downtown city area.

[0032] In particular embodiments, a dock **200** that is connected to an external power source (e.g., electrical grid) may not be configured with a solar panel, instead configured to rely on the external power source to operate its internal electrically-powered components. In other embodiments, a dock **200** may be configured with solar panel(s) to work in parallel with the electrical grid, or as a back-up power source to the electrical grid. In some embodiments, a dock **200** that is connected to an external power source may still be configured with a solar panel used to provide the necessary power for its electrically-powered components, while using power from the external power source to charge an electric rideable connected to the dock **200**.

[0033] In particular embodiments, each of the docking stations in a region, such as those illustrated in FIG. **9**, may communicate with each other and/or a server via a cloud. For example, each self-sufficient dock **200** may comprise a communication module that allows it to communicate to a server.

[0034] In particular embodiments, when a user requests a rideable from a docking station, a server may select a particular rideable docked at a station based on the user's origination location, destination location (e.g., specified by a user), estimated duration of the trip, user's preference of a rideable (e.g., electrically-powered rideable or not electrically-powered rideable), and/or the amount of charge left on the electrically-powered rideables. For example, if a user requests an electrically-powered rideable, a server may analyze the amount of charge left on each of the docked rideables and select one that is sufficiently charged. If a user requests an electrically-powered rideable at a docking station without charging capability and the docking station at the destination location is capable of charging, a server may select an electrically-powered rideable that has low charge but is still capable of reaching the destination location so the rideable can be charged when the user finishes the ride. In some embodiments, when a user requests a rideable from a docking station, a server may provide a user with all of the rideables that are available at the docking station to allow the user to select any particular rideable that is available and reserve that rideable for the user. The reserved rideable may then be held unavailable for a maximum time period (e.g., 5-30 minutes) before being made available to other users.

[0035] FIG. **10** illustrates an example method **1000** for communicating with individual docks at a docking station to allow a user to reserve a particular vehicle that is docked at a particular dock. In some embodiments, the method **1000** may be executed by a transportation management system that is in communication with various docking stations and users. The method may begin at step **1001** by receiving, from a client device associated with a user, a request to reserve a vehicle at a docking station. At step **1002**, the method may continue by identifying one or more vehicles docked at the docking station that are available by, for each of the one or more vehicles, communicating with a communication module carried by a housing structure to which the vehicle is docked. At step **1003**, the method may continue by sending information to the client device that identifies the one or more vehicles docked at the docking station that are available. At step **1004**, the method may continue by receiving, from the client device, a selection of a particular vehicle of the one or more vehicles docked at the docking station. At step **1005**, the method may continue by sending, to a communication module carried by a housing structure to which the particular vehicle is docked, instructions for reserving the particular vehicle for the user. Particular embodiments may repeat one or more steps of the method of FIG. **10**, where appropriate. Although this disclosure describes and

illustrates particular steps of the method of FIG. **10** as occurring in a particular order, this disclosure contemplates any suitable steps of the method of FIG. **10** occurring in any suitable order. Moreover, although this disclosure describes and illustrates an example method for communicating with individual docks at a docking station to allow a user to reserve a particular vehicle that is docked at a particular dock, this disclosure contemplates any suitable method for communicating with individual docks at a docking station to allow a user to reserve a particular vehicle that is docked at a particular dock, which may include all, some, or none of the steps of the method of FIG. **10**, where appropriate. Furthermore, although this disclosure describes and illustrates particular components, devices, or systems carrying out particular steps of the method of FIG. **10**, this disclosure contemplates any suitable combination of any suitable components, devices, or systems carrying out any suitable steps of the method of FIG. **10**.

[0036] FIG. **11** illustrates an example computer system **1100**. In particular embodiments, one or more computer systems **1100** perform one or more steps of one or more methods described or illustrated herein. In particular embodiments, one or more computer systems **1100** provide the functionalities described or illustrated herein. In particular embodiments, software running on one or more computer systems **1100** performs one or more steps of one or more methods described or illustrated herein or provides the functionalities described or illustrated herein. Particular embodiments include one or more portions of one or more computer systems **1100**. Herein, a reference to a computer system may encompass a computing device, and vice versa, where appropriate. Moreover, a reference to a computer system may encompass one or more computer systems, where appropriate.

[0037] This disclosure contemplates any suitable number of computer systems **1100**. This disclosure contemplates computer system **1100** taking any suitable physical form. As example and not by way of limitation, computer system **1100** may be an embedded computer system, a system-on-chip (SOC), a single-board computer system (SBC) (such as, for example, a computer-on-module (COM) or system-on-module (SOM)), a desktop computer system, a laptop or notebook computer system, an interactive kiosk, a mainframe, a mesh of computer systems, a mobile telephone, a personal digital assistant (PDA), a server, a tablet computer system, an augmented/virtual reality device, or a combination of two or more of these. Where appropriate, computer system **1100** may include one or more computer systems **1100**; be unitary or distributed; span multiple locations; span multiple machines; span multiple data centers; or reside in a cloud, which may include one or more cloud components in one or more networks. Where appropriate, one or more computer systems **1100** may perform without substantial spatial or temporal limitation one or more steps of one or more methods described or illustrated herein. As an example and not by way of limitation, one or more computer systems **1100** may perform in real time or in batch mode one or more steps of one or more methods described or illustrated herein. One or more computer systems **1100** may perform at different times or at different locations one or more steps of one or more methods described or illustrated herein, where appropriate.

[0038] In particular embodiments, computer system **1100** includes a processor **1102**, memory **1104**, storage **1106**, an input/output (I/O) interface **1108**, a communication interface **1110**, and a bus **1112**. Although this disclosure describes and illustrates a particular computer system having a particular number of particular components in a particular arrangement, this disclosure contemplates any suitable computer system having any suitable number of any suitable components in any suitable arrangement.

[0039] In particular embodiments, processor **1102** includes hardware for executing instructions, such as those making up a computer program. As an example and not by way of limitation, to execute instructions, processor **1102** may retrieve (or fetch) the instructions from an internal register, an internal cache, memory **1104**, or storage **1106**; decode and execute them; and then write one or more results to an internal register, an internal cache, memory **1104**, or storage **1106**. In particular embodiments, processor **1102** may include one or more internal caches for data,

instructions, or addresses. This disclosure contemplates processor **1102** including any suitable number of any suitable internal caches, where appropriate. As an example and not by way of limitation, processor **1102** may include one or more instruction caches, one or more data caches, and one or more translation lookaside buffers (TLBs). Instructions in the instruction caches may be copies of instructions in memory **1104** or storage **1106**, and the instruction caches may speed up retrieval of those instructions by processor **1102**. Data in the data caches may be copies of data in memory **1104** or storage **1106** that are to be operated on by computer instructions; the results of previous instructions executed by processor **1102** that are accessible to subsequent instructions or for writing to memory **1104** or storage **1106**; or any other suitable data. The data caches may speed up read or write operations by processor **1102**. The TLBs may speed up virtual-address translation for processor **1102**. In particular embodiments, processor **1102** may include one or more internal registers for data, instructions, or addresses. This disclosure contemplates processor **1102** including any suitable number of any suitable internal registers, where appropriate. Where appropriate, processor **1102** may include one or more arithmetic logic units (ALUs), be a multi-core processor, or include one or more processors **1102**. Although this disclosure describes and illustrates a particular processor, this disclosure contemplates any suitable processor.

[0040] In particular embodiments, memory **1104** includes main memory for storing instructions for processor **1102** to execute or data for processor **1102** to operate on. As an example and not by way of limitation, computer system **1100** may load instructions from storage **1106** or another source (such as another computer system **1100**) to memory **1104**. Processor **1102** may then load the instructions from memory **1104** to an internal register or internal cache. To execute the instructions, processor **1102** may retrieve the instructions from the internal register or internal cache and decode them. During or after execution of the instructions, processor **1102** may write one or more results (which may be intermediate or final results) to the internal register or internal cache. Processor **1102** may then write one or more of those results to memory **1104**. In particular embodiments, processor **1102** executes only instructions in one or more internal registers or internal caches or in memory **1104** (as opposed to storage **1106** or elsewhere) and operates only on data in one or more internal registers or internal caches or in memory **1104** (as opposed to storage **1106** or elsewhere). One or more memory buses (which may each include an address bus and a data bus) may couple processor **1102** to memory **1104**. Bus **1112** may include one or more memory buses, as described in further detail below. In particular embodiments, one or more memory management units (MMUs) reside between processor **1102** and memory **1104** and facilitate accesses to memory **1104** requested by processor **1102**. In particular embodiments, memory **1104** includes random access memory (RAM). This RAM may be volatile memory, where appropriate. Where appropriate, this RAM may be dynamic RAM (DRAM) or static RAM (SRAM). Moreover, where appropriate, this RAM may be single-ported or multi-ported RAM. This disclosure contemplates any suitable RAM. Memory **1104** may include one or more memories **1104**, where appropriate. Although this disclosure describes and illustrates particular memory, this disclosure contemplates any suitable memory.

[0041] In particular embodiments, storage **1106** includes mass storage for data or instructions. As an example and not by way of limitation, storage **1106** may include a hard disk drive (HDD), a floppy disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus (USB) drive or a combination of two or more of these. Storage **1106** may include removable or non-removable (or fixed) media, where appropriate. Storage **1106** may be internal or external to computer system **1100**, where appropriate. In particular embodiments, storage **1106** is non-volatile, solid-state memory. In particular embodiments, storage **1106** includes read-only memory (ROM). Where appropriate, this ROM may be mask-programmed ROM, programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), electrically alterable ROM (EAROM), or flash memory or a combination of two or more of these. This disclosure contemplates mass storage **1106** taking any suitable physical form. Storage **1106** may include one or more storage control units facilitating communication between

processor **1102** and storage **1106**, where appropriate. Where appropriate, storage **1106** may include one or more storages **1106**. Although this disclosure describes and illustrates particular storage, this disclosure contemplates any suitable storage.

[0042] In particular embodiments, I/O interface **1108** includes hardware, software, or both, providing one or more interfaces for communication between computer system **1100** and one or more I/O devices. Computer system **1100** may include one or more of these I/O devices, where appropriate. One or more of these I/O devices may enable communication between a person and computer system **1100**. As an example and not by way of limitation, an I/O device may include a keyboard, keypad, microphone, monitor, mouse, printer, scanner, speaker, still camera, stylus, tablet, touch screen, trackball, video camera, another suitable I/O device or a combination of two or more of these. An I/O device may include one or more sensors. This disclosure contemplates any suitable I/O devices and any suitable I/O interfaces **1108** for them. Where appropriate, I/O interface **1108** may include one or more device or software drivers enabling processor **1102** to drive one or more of these I/O devices. I/O interface **1108** may include one or more I/O interfaces **1108**, where appropriate. Although this disclosure describes and illustrates a particular I/O interface, this disclosure contemplates any suitable I/O interface.

[0043] In particular embodiments, communication interface **1110** includes hardware, software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) between computer system **1100** and one or more other computer systems **1100** or one or more networks. As an example and not by way of limitation, communication interface **1110** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or any other wire-based network or a wireless NIC (WNIC) or wireless adapter for communicating with a wireless network, such as a WI-FI network. This disclosure contemplates any suitable network and any suitable communication interface **1110** for it. As an example and not by way of limitation, computer system **1100** may communicate with an ad hoc network, a personal area network (PAN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), or one or more portions of the Internet or a combination of two or more of these. One or more portions of one or more of these networks may be wired or wireless. As an example, computer system **1100** may communicate with a wireless PAN (WPAN) (such as, for example, a Bluetooth WPAN), a WI-FI network, a WI-MAX network, a cellular telephone network (such as, for example, a Global System for Mobile Communications (GSM) network), or any other suitable wireless network or a combination of two or more of these. Computer system **1100** may include any suitable communication interface **1110** for any of these networks, where appropriate.

Communication interface **1110** may include one or more communication interfaces **1110**, where appropriate. Although this disclosure describes and illustrates a particular communication interface, this disclosure contemplates any suitable communication interface.

[0044] In particular embodiments, bus **1112** includes hardware, software, or both coupling components of computer system **1100** to each other. As an example and not by way of limitation, bus **1112** may include an Accelerated Graphics Port (AGP) or any other graphics bus, an Enhanced Industry Standard Architecture (EISA) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an Industry Standard Architecture (ISA) bus, an INFINIBAND interconnect, a low-pin-count (LPC) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (PCIe) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local (VLB) bus, or another suitable bus or a combination of two or more of these. Bus **1112** may include one or more buses **1112**, where appropriate. Although this disclosure describes and illustrates a particular bus, this disclosure contemplates any suitable bus or interconnect.

[0045] Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other types of integrated circuits (ICs) (such, as for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs),

hybrid hard drives (HHDs), optical discs, optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, solid-state drives (SSDs), RAM-drives, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, non-volatile, or a combination of volatile and non-volatile, where appropriate.

[0046] Herein, “or” is inclusive and not exclusive, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A or B” means “A, B, or both,” unless expressly indicated otherwise or indicated otherwise by context. Moreover, “and” is both joint and several, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, “A and B” means “A and B, jointly or severally,” unless expressly indicated otherwise or indicated otherwise by context.

[0047] The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes and illustrates respective embodiments herein as including particular components, elements, feature, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, features, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend. Furthermore, reference in the appended claims to an apparatus or system or a component of an apparatus or system being adapted to, arranged to, capable of, configured to, enabled to, operable to, or operative to perform a particular function encompasses that apparatus, system, component, whether or not it or that particular function is activated, turned on, or unlocked, as long as that apparatus, system, or component is so adapted, arranged, capable, configured, enabled, operable, or operative. Additionally, although this disclosure describes or illustrates particular embodiments as providing particular advantages, particular embodiments may provide none, some, or all of these advantages.

Claims

1. An apparatus for docking a vehicle, comprising: a housing structure comprising two side portions and a top portion connecting the two side portions, the two side portions and the top portion defining a cavity for receiving a portion of the vehicle; an electrically-powered locking assembly, coupled to the housing structure, for securing the housing structure to the vehicle subsequent to the portion of the vehicle being received by the cavity of the housing structure, wherein the portion of the vehicle being received by the cavity of the housing structure is a locking member of the vehicle; an electrically-powered communication device, carried by the housing structure, for enabling data communication over a network to indicate a status of the vehicle relative to the housing structure; and a solar panel, coupled to an external surface of one of the side portions of the housing structure, for independently converting light received by the solar panel into electrical energy to independently power (i) the electrically-powered locking assembly to unlock the vehicle from the housing structure, and (ii) the electrically-powered communication device to indicate the status of the vehicle.
